Benjamin Wu CV

National Astronomical Observatory of Japan
Mitaka, Tokyo, 181-8588, Japan
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https://wubenjamin.github.io/

Research Interests
· high performance computing · numerical simulations · big data visualization · machine learning ·
· computational fluid dynamics · star formation · space exploration · rover mapping and localization ·

Education

University of Florida, Gainesville, FL
Ph.D., Physics w/ minor in Astronomy — Dec 2012 – Aug 2016
Thesis: *Giant Molecular Cloud Collisions as Triggers of Star Formation*
Advisor: Dr. Jonathan C. Tan
M.S., Physics — Aug 2010 – Dec 2012

Duke University, Durham, NC
B.S., Mathematics and Physics — Aug 2005 – May 2009

Research Experience

National Astronomical Observatory of Japan, Tokyo, Japan
ALMA Division
Project Research Fellow w/ Dr. Patricio Sanhueza — Oct 2018 – Present

NASA Ames Research Center & SETI Institute, Mountain View, CA
Frontier Development Lab
Space Science & Machine Learning Researcher — Jun 2018 – Aug 2018

National Astronomical Observatory of Japan, Tokyo, Japan
Division of Theoretical Astronomy
Project Research Fellow w/ Dr. Fumitaka Nakamura — Nov 2017 – Jun 2018
JSPS Postdoctoral Fellow — Sep 2016 – Sep 2017

University of Florida, Gainesville, FL
Theoretical Astrophysics Group
Graduate Fellow w/ Dr. Jonathan Tan — Apr 2011 – Dec 2016

Laser Interferometer Gravitational Wave Observatory (LIGO)
Research Assistant w/ Dr. Guido Mueller, Dr. David Reitze, Dr. David Tanner — Sep 2009 – May 2011

Duke University, Durham, NC
Triangle Universities Nuclear Laboratories
Research Assistant w/ Dr. Henry R. Weller — May 2008 – May 2009

High Energy Physics Neutrino Group
Research Assistant w/ Dr. Kate Scholberg — May 2007 – Jul 2007

University of Central Florida, Orlando, FL
The College of Optics and Photonics
Research Assistant w/ Dr. Shin-Tson Wu — Summers 2004 – 2007

Awards & Fellowships

NASA Frontier Development Lab Fellow	2018
European Commission Marie Sklodowska-Curie Actions Seal of Excellence	2018
Japan Society for the Promotion of Science Postdoctoral Fellowship	2016 – 2017
NASA Florida Space Grant Dissertation Improvement Fellowship	2015 – 2016
SOFIA Student Grant	2015
NSF East Asia and Pacific Summer Institutes Fellowship	2014
NAOJ Visiting Scholar Grant	2013
UF Graduate School Fellowship	2010 – 2014
SPIE Optics and Photonics Educational Scholarship	2008

Refereed Publications

14. Bisbas, T., Tan, J. C., Csengeri, T., **Wu, B.,** Lim, W., Caselli, P., Gusten, R., Ricken, O., & Riquelme, D., "The Inception of Star Cluster Formation Revealed by [CII] Emission Around an Infrared Dark Cloud", Monthly Notices of the Royal Astronomical Society, 487, 54, (Jul 2018). [*NASA press release*]
13. **Wu, B.**, Tan, J.C., Nakamura, F., Christie, D., & Li, Q., "GMC Collisions as Triggers of Star Formation. VI. Collision-Induced Turbulence", Publications of the Astronomical Society of Japan, 70S, 57. (May 2018)
12. Li, Q., Tan, J. C., Christie, D., Bisbas, T., & **Wu, B.,** "The Interstellar Medium and Star Formation of Disk Galaxies. I. ISM and GMC Properties with Diffuse FUV Feedback", Publications of the Astronomical Society of Japan, 70S, 56. (May 2018)
11. Bisbas, T., Tanaka, K. E. I., Tan, J. C., **Wu, B.**, & Nakamura, F., "GMC Collisions as Triggers of Star Formation. V. Observational Signatures", The Astrophysical Journal, 850, 23-33 (Nov. 2017)
10. Christie, D., **Wu, B.**, & Tan, J. C., "GMC Collisions as Triggers of Star Formation. IV. The Role of Ambipolar Diffusion", The Astrophysical Journal, 848, 50-64 (Oct. 2017)
9. **Wu, B.**, Tan, J. C., Christie, D., Nakamura, F., Van Loo, S., & Collins, D., "GMC Collisions as Triggers of Star Formation. III. Density and Magnetically Regulated Star Formation", The Astrophysical Journal, 841, 88-106 (Jun. 2017).
8. **Wu, B.**, Tan, J. C., Nakamura, F., Van Loo, S., Christie, D., & Collins, D., "GMC Collisions as Triggers of Star Formation. II. 3D Turbulent Magnetized Simulations", The Astrophysical Journal, 835, 137-160, (Feb. 2017). [*featured in AstroBites*]
7. **Wu, B.**, Van Loo, S., Tan, J. C., & Bruderer, S., "GMC Collisions as Triggers of Star Formation. I. Parameter space exploration with 2D simulations", The Astrophysical Journal, 811, 56-76, (Sep. 2015).
6. Ren, H., Fox, D., **Wu, B.**, & Wu, S. T., "Liquid crystal lens with large focal length tunability and low operating voltage", Optics Express, 15, 11328-11335 (Sep. 2007).
5. Ren, H., Fox, D., Anderson, P. A., **Wu, B.**, & Wu, S. T., "Tunable-focus liquid lens controlled using a servo motor", Optics Express, 14, 8031-8036 (Sep. 2006).
4. Gauza, S., Wen, C. H., **Wu, B.**, & Wu, S. T., "Fast-response nematic liquid crystal mixtures", Journal of the Society of Information Display, 14, 241-246 (Mar. 2006).
3. Gauza, S., Wen, C. H., **Wu, B.**, & Wu, S. T., "High figure-of-merit nematic mixtures based on totally unsaturated isothiocyanate liquid crystals", Liquid Crystals, 33, 705-710 (Jun. 2006).
2. Wen, C. H., **Wu, B.**, Gauza, S., Nie, X., & Wu, S. T., "Dopant-enhanced vertical alignment of lateral fluorinated liquid crystals", Molecular Crystals and Liquid Crystals, 454, 315-324 (Jun. 2006).
1. Wen, C. H., **Wu, B.**, Gauza, S., & Wu, S. T., "Dopant-induced contrast enhancement of vertically aligned liquid crystal cells", Journal of Display Technology, 1, 234-239 (Dec. 2005).

Presentations

37. East Asian ALMA Science Workshop 2018, Osaka, Japan. Poster. (Dec 2018)
36. Workshop on Theoretical Astronomy, Kurashiki, Japan. Contributed. (Dec 2018)
35. NASA Ames SOFIA Colloquium, Mountain View, CA. Invited. (Sep 2018)
34. NASA Frontier Development Lab Event Horizon, Santa Clara, CA. (Aug 2018) [*featured in IEEE Spectrum*]
33. NASA Frontier Development Lab Final Technical Presentation, Mountain View, CA. (Aug 2018)
32. Workshop: Star Formation Triggered by Cloud-Cloud Collision, Nagoya, Japan. Invited. (May 2018)
31. High Mass Star Formation Workshop, Yamaguchi, Japan. Contributed. (Mar 2018)

30. Magnetic Fields or Turbulence: Which is the critical factor for the formation of stars and planetary disks?, Hsinchu, Taiwan. Contributed. (Feb 2018)
29. Division of Theoretical Astronomy Workshop, Mizusawa, Japan. Contributed. (Nov 2017)
28. Astronomical Society of Japan Fall Meeting, Sapporo, Japan. Contributed. (Sept 2017)
27. Star Formation in Different Environments: From Local Clouds to Distant Galaxies, Quy Nhon, Vietnam. Contributed. (Aug 2017)
26. Astronomical Society of Japan Spring Meeting, Fukuoka, Japan. Contributed. (Mar 2017)
25. NAOJ Theory Seminar, Mitaka, Tokyo, Japan. Invited. (Mar 2017)
24. Hiroshima Astrophysical Science Center, Hiroshima University, Hiroshima, Japan. Invited. (Mar 2017)
23. Research Center for Space and Cosmic Evolution, Ehime University, Matsuyama, Japan. Invited. (Feb 2017)
22. NAOJ ALMAJ Seminar, Mitaka, Tokyo, Japan. Invited. (Feb 2017)
21. University of Florida Department of Physics, Gainesville, FL. PhD Defense. (Jun 2016)
20. Conference; Star formation, Magnetic Fields, and Diffuse Matter in the Galaxy, Madison, WI. Poster. (May 2016)
19. From Stars to Massive Stars, Gainesville, FL. Contributed. (Apr 2016)
18. University of Florida Department of Physics Graduate Student Seminar, Gainesville, FL. Contributed. (Feb 2016)
17. American Astronomical Society 227th Meeting, Kissimmee, FL. Contributed Dissertation. (Jan 2016)
16. Astrophysical Simulation Workshop, Gainesville, FL. Invited. (Oct 2015)
15. Astronomy Graduate Student Symposium, Gainesville, FL. Contributed. (Sept 2015)
14. 30 Years of PhotoDissociation Regions, Asilomar, CA. Poster, honorable mention. (Jul 2015)
13. University of Florida Department of Physics, Gainesville, FL. PhD Candidacy. (Nov 2014)
12. Filamentary Structure in Molecular Clouds, Charlottesville, VA. Poster. (Nov 2014)
11. NAOJ Seminar, Mitaka, Tokyo, Japan. Invited. (Jul 2014)
10. JSPS Summer Program Symposium, Tokyo, Japan. Poster. (Jun 2013)
9. Impact of Galactic Structure on Star Formation, Sapporo, Japan. Contributed. (Feb 2014)
8. University of Florida Department of Astronomy Graduate Student Symposium, Gainesville, FL. Contributed. (Aug 2013)
7. NAOJ ALMAJ Seminar, Mitaka, Tokyo, Japan. Invited. (May 2013)
6. NAOJ Theory Seminar, Mitaka, Tokyo, Japan. Invited. (May 2013)
5. From Stars to Life, Gainesville, FL. Poster. (Apr 2013)
4. Florida Star & Planet Formation Day, Gainesville, FL. Contributed. (Feb 2013)
3. Star Formation & The Interstellar Medium, Berkeley, CA. Poster. (Nov 2012)
2. University of Florida Department of Astronomy Graduate Student Symposium, Gainesville, FL. Contributed. (Aug 2012)
1. Florida Star & Planet Formation Day, Gainesville, FL. Contributed. (Feb 2012)

Observing & Super-computing Grants

NOEMA: *Investigating the Kinematic Imprints of an Interstellar Collision.* (Co-I) 2018
JCMT: *Mapping Magnetic Field in Infrared Dark Clouds.* (Co-I) 2018
Nobeyama: *A statistical study of massive clumps in the early stages of cluster formation.* (Co-I) 2018
SMA: *G034.77-00.55 and the first fully resolved CJ-type interstellar shock.* (Co-I) 2018
JCMT: *Mapping Magnetic Field in Infrared Dark Clouds.* (Co-I) 2017
SOFIA: *The Inception of Star Cluster Formation: [CII] emission from IRDCs – II. Massive Protoclusters and Testing GMC Collision Models for Triggered Star Formation.* (Co-I) 2017
CfCA Allocation: *Evolution of Star Clusters Triggered by GMC Collisions*. (PI) 2017
ALMA: *Magnetic Fields in High-Mass Star Formation*. (Co-I) 2017
XSEDE Allocation: *Simulations of Star Cluster Formation Triggered by GMC Collisions*. (Co-I) 2016
SOFIA: *The Inception of Star Cluster Formation: [CII] emission from IRDCs*. (Co-I) 2015

Computing Skills & Information

Programming: Python, C++, Fortran, Bash, XHTML/CSS, MPI
Software Tools: Git/Mercurial, LaTeX, VisIt, MATLAB, Mathematica/Maple, Office
Machine Learning: TensorFlow, Google Cloud Platform, Convolutional Neural Networks
Environment: Linux (server admin), High Performance Computing (PBS/Slurm), macOS, Windows
Languages: English (native), Mandarin (intermediate), Spanish (basic), Japanese (basic)
Citizenship: U.S.A.

Teaching

Recitation Lecturer, UF Department of Physics
PHY 2049 – Physics with Calculus II (Electricity & Magnetism) 2014
PHY 2048 – Physics with Calculus I (Mechanics, Kinematics, Fluids) 2013 – 2014

- Taught physics in a classroom setting (total 9 sections of ~30 students each)
- Prepared lectures; reviewed problem-solving techniques; wrote weekly quizzes; held course-wide review sessions
- Excellent instructor rating of 4.6/5.0 (department average: 3.9/5.0)

Lab Instructor, UF Department of Physics
PHY 2004L/2048L/2053L – Physics I Labs 2011 – 2013

- Interactively led and troubleshot hands-on laboratory experiments (total 8 sections of ~16 students each)

Tutor, Duke University
Peer Tutoring Program: Physics I, Physics II, Calculus 2007 – 2009
Physics Help Room: Physics I, Physics II 2007 – 2009

- Tutored undergraduate peers in physics and calculus courses in one-on-one and group settings

Student Co-Advising

Takayuki Ohgami (graduate student, Yamaguchi University) 2016 – 2017
Qi Li (graduate student, University of Florida) 2015 – 2016
Eric B. Deleeuw (undergraduate student, University of Florida) 2009 – 2010

Outreach

Mitaka Open House Day, National Astronomical Observatory of Japan 2016 – 2018
Assisted and maintained the 3D planetarium for annual event highlighting space science research and education in Japan, targeting the greater Tokyo area (~2000 attendees)

After Hours Adventure: Star Party, Kennedy Space Center 2016
Set up interactive educational stations and stargazing activities for a family-friendly event exploring astronomy and space technology themes (~500 attendees)

Springs and Stars, various Florida state parks 2016
Provided inspirational, interactive, and educational astronomy outreach and stargazing activities targeting underprivileged students in rural Florida areas (~120 attendees)

Starry Night, Florida Museum of Natural History 2011 – 2015
Built and managed interactive activity stations in annual event showcasing a wide array of topics in astrophysics and astronomy, designed for the general audience (~2000 attendees)

References

Dr. Jonathan C. Tan
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Department of Space, Earth, and Environment
Chalmers University of Technology, Göteborg, Sweden &
Department of Astronomy,
University of Virginia, Charlottesville, VA, USA
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Dr. Sven Van Loo
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University of Leeds, Leeds, UK
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